MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSEPSY200 - BIOPSYCHOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question- Choose the most appropriate answer

Answer all questions

Each question carries 1 marks

1. Which of the following provides the most detailed three-dimensional view of the structure of the living human brain?

Answer: iv. MRI

2.The midbrain is a part of which larger brain division?

Answer: iii. Brainstem

3. During maximum muscle contraction, EMG shows:

Answer: i. Dense interference pattern

4. Recurrent binge-eating without compensatory behaviour is characteristic of:

Answer: ii. Binge-eating disorder

5. Which process is responsible for the removal of neurotransmitters from the synaptic cleft by nearby glial cells?

Answer: iii. Reuptake

6. Which of the following is an example of a violation of ethical research practice in biopsychology?

Answer: iii. Fabricating data to support a hypothesis

7. Damage to the cerebellum is most likely to cause

Answer: ii. Difficulty in balance and coordination.

8. What is the name of the theory that proposes that testosterone is converted to estradiol in the brain to facilitate sexual differentiation?

Answer: i. Aromatization Hypothesis

9. What does electrical stimulation of the brain typically cause?

Answer: ii.Temporary activation of neurons

10. Which hypothalamic area is mainly responsible for triggering thirst?

Answer: i. Lateral Hypothalamus

11. Electrostatic pressure refers to:

Answer: ii. The attraction and repulsion among charged ions

12. The HPA axis primarily regulates:

Answer: iii. Stress Responses

13. The primary hormones secreted by the adrenal medulla during stress are:

Answer: iii. Epinephrine and norepinephrine

14. A postsynaptic potential that makes the postsynaptic neuron less likely to fire an action potential.

Answer: iv. Inhibitory postsynaptic potential (IPSP)

15. Select the metabolic condition most strongly linked with insulin resistance and memory impairment.

Answer: iv. Diabetes

16. The nervous system responsible for the voluntary control of skeletal muscles is

Answer: ii. Somatic nervous system

17. What is the Papez circuit related to?

Answer: ii. Emotional expression and memory

18. Insulin, a hormone that regulates blood glucose levels, is secreted by which gland?

Answer: Errors in options. Give one mark. The correct answer is pancreas

19. Recognise the process of neurotransmitter release

Answer: i. Exocytosis

20. Positive feedback is beneficial in:

Answer: ii. Childbirth and lactation

Part B

Short Answer type questions

Answer any 10 questions

Each question carries 3 marks

21. What is a reflex arc? Name its main components.

A reflex arc is a neural pathway that controls a reflex action. It is an automatic, rapid response to a specific stimulus that bypasses the brain. The main Components are

- Sensory Receptor: Detects the stimulus.
- **Sensory (Afferent) Neuron**: Carries information to the CNS.
- **Integration Centre (Interneuron)**: Processes the information.
- **Motor (Efferent) Neuron**: Carries the response command away from the CNS.
- Effector (Muscle/Gland): Performs the reflexive action.

22. Describe neuropathology and explain its role in understanding brain disorders.

Neuropathology is the branch of neuroscience and pathology that studies diseases of the nervous system—particularly the brain, spinal cord, and peripheral nerves—by examining tissue samples. Role - Identifying Disease Mechanisms, Linking Structure to Function, Improving Diagnosis and Classification, Guiding Treatment and Research, Predicting Prognosis.

23. Explain some potential benefits of endorphin release.

Endorphins are endogenous opioids that act as natural painkillers and mood elevators. The benefits are

- Pain Reduction: They inhibit the transmission of pain signals in the brain and spinal cord, providing analgesia.
- Stress and Anxiety Relief: They produce feelings of well-being and euphoria, helping the body cope with stress ("runner's high").
- Modulation of Appetite and Release of Sex Hormones: They influence other physiological processes linked to reward and homeostasis.

24. What is the significance of the limbic system in behaviour?

The limbic system is a group of interconnected brain structures crucial for emotion, motivation, learning, and memory. The Significance are

- Amygdala: Central to emotional responses, especially fear and aggression.
- Hippocampus: Vital for forming new long-term memories (memory consolidation).
- Hypothalamus: Regulates motivated behaviors (fighting, fleeing, feeding, mating) via hormonal and autonomic outputs.

Together, they link higher cognitive processes with primitive emotional drives, allowing experiences to be emotionally colored and remembered.

25. What is the primary role of cortisol in the stress response?

Cortisol is the primary glucocorticoid hormone released by the adrenal cortex during stress. The primary role is to mobilize energy and suppress non-essential functions to help the body cope with a stressor.

- It increases blood sugar (gluconeogenesis).
- It enhances the brain's use of glucose.

- It enabling fight-or-flight responses.
- It modulates inflammatory and immune responses.

26. Define hemispheres. Mention the functions of the left and right hemispheres.

The cerebral cortex is divided into two halves, the left and right cerebral hemispheres, connected by the corpus callosum.

- Left Hemisphere Functions: Dominant for language (Broca's and Wernicke's areas), logical reasoning, analytical thinking, and sequential processing.
- Right Hemisphere Functions: Dominant for spatial awareness, face recognition, visual imagery, music perception, and holistic processing.

27. Describe the role of acetylcholine.

Acetylcholine (ACh) is a small-molecule neurotransmitter, a critical neurotransmitter with multiple roles:

- At the Neuromuscular Junction: It is the excitatory transmitter that triggers muscle contraction.
- In the Brain: It is vital for attention, learning, memory, and arousal. The degeneration of ACh-releasing neurons is a key feature of Alzheimer's disease.
- In the Autonomic Nervous System: It is the main transmitter for pre-ganglionic neurons in both the sympathetic and parasympathetic divisions, and for post-ganglionic neurons in the parasympathetic division.

28. Describe how growth hormone affects learning.

Growth hormone (GH) plays an important role in brain development and learning. It promotes the growth and repair of brain cells and enhances synaptic plasticity, which is essential for learning and memory formation. GH also increases the production of insulin-like growth factor-1 (IGF-1), which supports neural growth, cognitive function, and overall brain performance.

29. Explain satiation and satiety centre.

These are hypothalamic centres that regulate eating behavior.

- Satiety Centre: The Ventromedial Hypothalamus (VMH). When stimulated, it signals "fullness" and stops eating. Damage leads to hyperphagia (overeating).

- Satiation Centre (Hunger Centre): The Lateral Hypothalamus (LH). When stimulated, it initiates eating. Damage leads to aphagia (lack of eating). These centers integrate signals like leptin and ghrelin to control food intake.

30. Write a short note on neurotransmitters used in sympathetic and parasympathetic systems.

Sympathetic Nervous System - fight-or-flight

- Pre-ganglionic neuron: Releases Acetylcholine (ACh) onto nicotinic receptors.
- Post-ganglionic neuron: Primarily releases Norepinephrine (NE) onto adrenergic receptors on target organs.

Parasympathetic Nervous System ("Rest-and-Digest"):

- Pre-ganglionic neuron: Releases Acetylcholine (ACh) onto nicotinic receptors.
- Post-ganglionic neuron: Releases Acetylcholine (ACh) onto muscarinic receptors on target organs.

31. Explain the role of the amygdala in processing sexually arousing stimuli.

The amygdala is a key limbic structure for processing emotional significance, including sexual arousal.

- It evaluates sensory stimuli (visual, olfactory) for their potential sexual and rewarding content.
- It assigns emotional value to these stimuli, triggering autonomic (e.g., increased heart rate) and hormonal responses.
- It modulates motivation and approach behaviors by interacting with the hypothalamus and other reward-related areas, facilitating sexual behavior.

32. **Describe how the influx of sodium (Na^+) and efflux of potassium (K^+) ions work together during the action potential.**

- Depolarization (Rising Phase): A stimulus opens voltage-gated sodium channels. Na^+ rushes into the neuron down its electrochemical gradient, causing a rapid positive shift in the membrane potential.
- Repolarization (Falling Phase): Sodium channels inactivate. Voltage-gated potassium channels open. K^+ rushes out of the neuron, down its concentration gradient, restoring the negative internal charge.

Together, this coordinated influx of Na^+ and efflux of K^+ creates the rapid rise and fall of the action potential, allowing it to propagate along the axon.

33. Discuss the role of Broca's area in speech and word formation.

Broca's area, located in the left frontal lobe, is critical for the motor aspects of speech production. It is responsible for planning the complex motor sequences required for articulate speech. It sends instructions to the motor cortex to coordinate the muscles of the lips, tongue, jaw, and larynx. Result of Damage (Broca's Aphasia): Individuals understand language but have great difficulty producing it. Their speech is non-fluent, effortful, and telegraphic (e.g., "Walk... dog... today"), but the meaning is often preserved.

Part C

Long Answer type questions

Answer any 2 questions

Each question carries 10 marks

34. Explain how hormones influence sexual behaviour

Hormones play a major role in shaping and regulating sexual behaviour through two main effects – organizational and activational effects. These work together to form the biological basis of sexual motivation and activity.

- **Organizational Effects** (Permanent Brain Structuring) - These are long-lasting effects that occur during early development (prenatal or early postnatal period). They help in “organizing” the brain into a male or female pattern. Testosterone is the key hormone. In males, testosterone is converted into estradiol in the brain, which masculinizes and defeminizes the brain.
- **Activational Effects** (Temporary Activation in Adulthood) - These are short-term and reversible effects that occur after puberty. Hormones activate the neural circuits formed earlier, controlling sexual desire and behaviour. In females: Estrogen and progesterone levels rise around ovulation, increasing sexual receptivity. In males: Testosterone maintains sex drive, erectile function, and sperm production. Removal of testosterone reduces sexual activity, but it can be restored with hormone therapy.

- Puberty and Hormonal Regulation - During puberty, a surge in sex hormones activates sexual motivation and secondary sexual characteristics. The hypothalamic–pituitary–gonadal (HPG) axis controls hormone release and reproductive function.
- Brain Areas and Neurotransmitters - Hormones act on brain regions like the hypothalamus, amygdala, and preoptic area, influencing arousal and partner preference. They interact with dopamine (pleasure), serotonin, and oxytocin (bonding and intimacy).
- Psychological and Environmental Influence - While hormones set the biological foundation, sexual behaviour is also shaped by learning, emotions, relationships, and cultural factors.

35. Describe the structure of the forebrain and explain how each of its major components contributes to higher cognitive and physiological functions.

The forebrain (or prosencephalon) is the largest and most rostral part of the brain, responsible for the most complex cognitive, sensory, and motor functions. It can be divided into two main parts: the Telencephalon and the Diencephalon.

- ✓ Telencephalon

1. Cerebral Cortex: The convoluted outer layer of the brain.

2. Lobes

- Frontal Lobe: Executive functions (planning, decision-making, working memory), voluntary motor control (Primary Motor Cortex), and speech production (Broca's Area)
- Parietal Lobe: Somatosensory processing (touch, pressure, pain - Primary Somatosensory Cortex), spatial awareness, and navigation.
- Temporal Lobe: Auditory processing (Primary Auditory Cortex), complex visual perception (face recognition), and memory formation (Hippocampus, which is medial to the lobe).
- Occipital Lobe: Visual processing (Primary Visual Cortex).

- ✓ Diencephalon
  Located beneath the cerebral hemispheres, it serves as a central relay and control station.

1. Thalamus:

- Structure: A paired, egg-shaped structure.
- Function: The "Gateway to the Cortex." It processes and relays almost all sensory information (except smell) to the appropriate primary sensory areas of the cerebral cortex. It also plays a role in regulating consciousness, sleep, and alertness.

2. Hypothalamus:

- Structure: A small but vital structure below the thalamus.
- Function: The master regulator of homeostasis and motivated behaviors. It controls:
  - Autonomic Nervous System: Heart rate, digestion, etc.
  - Endocrine System: Via the pituitary gland, it regulates hormone release for stress, growth, and reproduction.
  - Motivated Behaviors: Feeding, drinking, fighting, fleeing, and mating.
  - Circadian Rhythms: Via the Suprachiasmatic Nucleus (SCN).

36. Structural Differences between the Male and Female Hypothalamus.

Hypothalamus. Sexual dimorphism in the hypothalamus underlies differences in reproductive physiology and behaviour. These structural differences are primarily caused by exposure to sex hormones during prenatal and early postnatal development (organizational effects).

1. Sexually Dimorphic Nucleus of the Preoptic Area (SDN-POA)

- Location: Preoptic Area of the hypothalamus.
- Difference: This nucleus is significantly larger (2-3 times) and contains more neurons in males than in females across many species, including humans (where it is called the INAH-3).

- Functional Significance: The SDN-POA is crucial for the expression of male typical sexual behaviour. Its larger size in males is directly organized by prenatal exposure to testosterone (converted to estradiol). Lesions in this area disrupt mounting behaviour in males.

2. Anterior Hypothalamic Nucleus

- Location: Anterior Hypothalamus.
- Difference: In some species, specific cell groups within this region are larger in males. In humans, a related area is often discussed.
- Functional Significance: Believed to be involved in gender identity and sexual orientation, though the research is complex and ongoing.

3. Vasopressinergic Pathways

- Difference: Males have a denser network of neurons producing vasopressin in specific hypothalamic areas (like the bed nucleus of the stria terminalis and medial amygdala) that project to the lateral septum.
- Functional Significance: Vasopressin in these circuits is associated with paternal behaviour, social bonding, and aggression—traits often more pronounced in males.

4. The GnRH Pulse Generator

- Location: Medial Basal Hypothalamus/Arcuate Nucleus.
- Difference: While the structure of the neurons is similar, the pattern of activity is profoundly different. In females, the pulse generator produces a cyclic (surge) release of Gonadotropin-Releasing Hormone (GnRH), leading to the ovulatory cycle. In males, it produces a steady, tonic release.
- Functional Significance: This difference is responsible for the cyclic nature of female reproduction (menstrual/estrous cycle) versus the acyclic, constant fertility in males. This "cycling center" is defeminized in males by prenatal testosterone.

37. Illustrate the structure of synapse in detail and list out the types of synapses.

A synapse is a junction with a minute gap that separates two neurons (nerve cells), the basic unit of the nervous system in the brain. Classification Based on Structures - Axo-dendritic Synapses, Axo-somatic Synapses, Axo-axonic Synapses, Dendo-dendritic Synapses. Classification Based on Signal Transduction Methods: Electrical Synapses, Chemical Synapses (Excitatory Chemical Synapses, Inhibitory Chemical Synapses). Structure of synapses - The typical synapse comprises three main elements: the presynaptic neuron, the synaptic cleft, and the postsynaptic neuron. Presynaptic Neuron: Synaptic Vesicles, Active Zone, Cytoskeletal Elements. Synaptic Cleft. Postsynaptic Neuron: Functions - Receptor Activation, Ion Flow Regulation: